

**FAST- National University of Computer and
Emerging Sciences**



Discrete Structures

Assignment # 2

SUBMITTED TO: Dr. Fehmida Usmani

SUBMITTED BY: Muhammad Moiz Ansari

SECTION: BS(CS) - F

Roll No: 23i - 0523

Assignment-2

①

Q-1:

a_n

$$a_n = 3n + 2$$

(i)

$$a_0 = 0 + 2 = 2$$

$$a_1 = 3 + 2 = 5$$

$$a_2 = 6 + 2 = 8$$

$$a_3 = 9 + 2 = 11$$

$$a_4 = 12 + 2 = 14$$

(ii)

$$\text{Series} = \underset{\substack{\uparrow \\ a_0}}{2} + 5 + 8 + 11 + 14 + \dots + (3n + 2)$$

$$a_0 = 2$$

$$a_n = 3n + 2$$

(Sum of first n terms starting with $a_0 = \frac{(n+1)(a_0 + a_n)}{2}$)

$$= \frac{(n+1)(2 + 3n + 2)}{2}$$

$$= \frac{(n+1)(3n+4)}{2}$$

$\therefore$ Its from a_0 , so for first n terms replace n with $(n-1)$

$$\Rightarrow = \frac{(n-1+1)[3(n-1)+4]}{2}$$

$$= \frac{n(3n-3+4)}{2}$$

$$\boxed{= \frac{3n^2 + n}{2}} \text{ Ans. } (n \geq 0)$$

(2)

(b)

Formula = ?

Seq: $0 + 2 + 4 + \dots + 2n$ ~~Sum of~~ $a_0 = 0$

$$a_n = 2n$$

Sum of first n terms from $a_0 = \frac{(n+1)(0+2n)}{2}$

$$= \frac{(n+1)(2n)}{2}$$

$$= n(n+1)$$

Take n as $(n-1)$ bcz starting from a_0 :

$$\text{Formula} = (n-1)(n-1+1)$$

$$= (n-1)n$$

$$\boxed{= n^2 - n} \quad \underline{\text{Ans.}} \quad (n \geq 0)$$

(c)

(i)

$$\sum_{k=1}^{10} (2k+1)$$

$$a_1 = 2(1)+1$$

$$= 3$$

$$a_n = a_{10} = 2(10)+1$$

$$= 21$$

$$n = 10$$

$$\Rightarrow \sum_{k=1}^{10} (2k+1) = \frac{n}{2} (a_1 + a_n)$$

$$= \frac{10}{2} (3+21)$$

$$\boxed{= 120} \quad \underline{\text{Ans.}}$$

(ii)

$$\prod_{k=1}^5 (k^2+1)$$

$$= (1^2+1)(2^2+1)(3^2+1)(4^2+1)(5^2+1)$$

$$= 2 \cdot 5 \cdot 10 \cdot 17 \cdot 26$$

$$\boxed{= 44200} \quad \underline{\text{Ans.}}$$

(d)

(i)

$$\begin{aligned} & \frac{10!}{8! \cdot 2!} \\ &= \frac{8! \cdot 9 \cdot 10}{8! \cdot (2 \cdot 1)} \\ &= \frac{90}{2} \\ &= 45 \text{ Ans.} \end{aligned}$$

(ii)

$$\sum_{k=0}^5 \frac{1}{k!}$$

A/c to Taylor series,

$$\begin{aligned} \sum_{k=0}^n \frac{1}{k!} &= e - R_n \\ &= e - \frac{e^c}{(n+1)!} \quad (\text{Here, } e^c = 1) \end{aligned}$$

$$= 2.718 - \frac{1}{(5+1)!}$$

$$= 2.718 - \frac{1}{720}$$

$$= 2.718 \text{ Ans.}$$

(e)

$$3, 6, 12, 24, 48, 96, 192, \dots$$

$$= 3, 3(2^1), 3(2^2), 3(2^3), \dots, \boxed{3(2^n)} \text{ Ans. } (n \geq 0).$$

Next three terms:

$$\begin{aligned} a_7 &= 3(2^7) = 384 \\ a_8 &= 3(2^8) = 768 \\ a_9 &= 3(2^9) = 1536 \end{aligned}$$

(4)

Q-2:

(a)

(i)

$$5 + 10 + 15 + 20 + \dots + 300 \text{ --- ①}$$

$$a_1 = 5$$

$$a_n = 300$$

For n:

$$\text{①} \Rightarrow 5(1) + 5(2) + 5(3) + 5(4) + \dots + 5(n) \text{ --- ②}$$

$$\text{① \& ②} \Rightarrow 5(n) = 300$$

$$n = \frac{300}{5} = 60$$

$$n = 60$$

$$\begin{aligned} \Rightarrow \text{Sum} &= \frac{n}{2} (a_1 + a_n) \\ &= \frac{60}{2} (5 + 300) \end{aligned}$$

$$\boxed{\text{Sum} = 9150} \text{ Ans}$$

(ii)

$$3 + 3^2 + 3^3 + \dots + 3^n$$

$$S_n = a_1 \frac{1-r^n}{1-r} \text{ --- ①}$$

$$a_1 = 3$$

$$r = \frac{3^2}{3} = 3$$

$$r = 3$$

$$\text{①} \Rightarrow S_n = 3 \left(\frac{1-3^n}{1-3} \right)$$

$$\boxed{S_n = -\frac{3}{2} (1-3^n)} \text{ Ans}$$

(iii)

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$$

$$a_1 = \frac{1}{2}$$

$$r = \frac{\frac{1}{4}}{\frac{1}{2}}$$

$$r = \frac{2}{2}$$

$$r = \frac{1}{2}$$

$$S_n = a_1 \left(\frac{1-r^n}{1-r} \right)$$

$$= \frac{1}{2} \left(\frac{1 - \left(\frac{1}{2}\right)^n}{1 - \frac{1}{2}} \right)$$

$$= \frac{1}{2} \left(\frac{1 - \frac{1}{2^n}}{\frac{2-1}{2}} \right)$$

$$= \frac{1}{2} \frac{\left(\frac{2^n-1}{2^n}\right)}{\left(\frac{1}{2}\right)}$$

$$S_n = \frac{2^n - 1}{2^n} \quad \underline{\text{Ans.}}$$

(b)

$$2 + 4 + 8 + \dots$$

$$2^1 + 2^2 + 2^3 + \dots + 2^n$$

n^{th} generation

$$n = 40$$

$$a_1 = 2$$

$$r = \frac{2^2}{2}$$

$$r = 2$$

$$S_n = S_{40} = a_1 \left(\frac{1-r^n}{1-r} \right)$$

$$= 2 \left(\frac{1-2^{40}}{1-2} \right)$$

$$= -2(1-2^{40})$$

$$S_{40} = 2 \cdot 19 \times 10^{12} \text{ ancestors} \quad \underline{\text{Ans.}}$$

⑥

(c)

To prove: $n^3 + 2n$ is divisible by 3, $n > 0$ or $n \in \mathbb{Z}^+$

• Basis step:

$$P(1) \Rightarrow (1)^3 + 2 = 1 + 2 = 3 \text{ is divisible by 3 } \checkmark$$

• Inductive Hypothesis: $(P(k) \rightarrow P(k+1))$

$$P(k) \Rightarrow k^3 + 2k \text{ is divisible by 3} \text{ --- ①}$$

To show: $P(k+1) \Rightarrow (k+1)^3 + 2(k+1)$ is div by 3

$$\begin{aligned} \Rightarrow (k+1)^3 + 2(k+1) &= k^3 + (1)^3 + 3k(k+1) + 2k + 2 \\ &= k^3 + 2k + 3k(k+1) + 3 \\ &= \underline{k^3 + 2k} + \underline{3(k^2 + k + 1)} \end{aligned}$$

① $\Rightarrow k^3 + 2k$ is div by 3, so this whole expression is div by 3, $\because 3$ is also divisible by 3.

Q-4:

(a)

$$V_1 = ? \quad \underline{V_k = V_{k-1} + V_{k-1} + 1}$$

$$V_2 = ?$$

$$V_3 = ?$$

$$V_4 = ?$$

$$\underline{V_1 = 1}, \underline{V_2 = 3}$$

$$\$ V_3 = V_2 + V_2 + 1$$

$$V_3 = 3 + 3 + 1$$

$$\boxed{V_3 = 7}$$

$$V_4 = V_3 + V_3 + 1$$

$$V_4 = 7 + 7 + 1$$

$$\boxed{V_4 = 15}$$

$$V_5 = V_4 + V_4 + 1$$

$$V_5 = 15 + 15 + 1$$

$$\boxed{V_5 = 31}$$

$$V_6 = V_5 + V_5 + 1$$

$$V_6 = 31 + 31 + 1$$

$$\boxed{V_6 = 63}$$

(b)

(i)

$$\boxed{a_1 = 2}$$

$$a_2 = 3(2) + 2 = 8$$

$$\boxed{a_2 = 8}$$

$$a_3 = 8 + 1 + 8 + 1 + 8 = 3(8) + 2$$

$$\boxed{a_3 = 26}$$

$$a_4 = 26 + 1 + 26 + 1 + 26 = 3(26) + 2$$

$$\boxed{a_4 = 80}$$

(ii)

From the above observation,

$a_n = 3(a_{n-1}) + 2$	<u>Aus.</u>
$a_0 = 0$	

(c)

To prove: $F_0^2 + F_1^2 + F_2^2 + \dots + F_n^2 = F_n \times F_{n+1}$

$$F_0 = 1$$

$$F_1 = 1$$

$$\Rightarrow 1^2 + 1^2 + 2^2 + 3^2 + 5^2 + \dots + F_n^2 = F_n \times F_{n+1}$$

$$\Rightarrow 1 + 1 + 4 + 9 + 25 + 64 + \dots + F_n^2 = F_n \times F_{n+1}$$

• Basis step:

$$\begin{aligned} \Rightarrow P(0) &\Rightarrow (F_0)^2 = (1)^2 \\ &= 1 = F_0 \times F_1 \\ &= 1 \times 1 \\ &= 1 \quad \checkmark \end{aligned}$$

⑧

• Inductive Step: $(P(k) \rightarrow P(k+1))$

Inductive hypothesis:

$$P(k) \Rightarrow 1+1+4+9+\dots+F_k^2 = F_k \times F_{k+1} \text{ --- ①}$$

To show: $P(k+1) \Rightarrow 1+1+4+9+\dots+(F_{k+1})^2 = F_{k+1} \times F_{k+2}$

$$\text{①} \Rightarrow 1+1+4+\dots+F_k^2 = F_k \times F_{k+1}$$

$$\Rightarrow (1+1+4+\dots+F_k^2) + (F_{k+1})^2 = (F_{k+1})(F_k) + (F_{k+1})^2$$

$$1+1+4+\dots+(F_{k+1})^2 = F_{k+1} (F_k + F_{k+1})$$

$$1+1+4+\dots+(F_{k+1})^2 = F_{k+1} \times F_{k+2} \text{ --- ② } (\because F_{k+2} = F_{k+1} + F_k)$$

② $\Rightarrow$ Hence, proved

(d)

Rate of increase = 3%

$P_0 = 50$ million (Initial population)

$P_{25} = ?$

$$\begin{aligned} P_1 &= P_0 + P_0 \times 3\% \\ &= P_0 + P_0 \times \frac{3}{100} \\ &= 50 + 50 \left(\frac{3}{100} \right) \end{aligned}$$

$$P_1 = \frac{103}{2}$$

$$\begin{aligned} P_2 &= P_1 + P_1 \times \frac{3}{100} \\ &= \frac{103}{2} + \frac{103}{2} \times \frac{3}{100} \end{aligned}$$

$$P_2 = \frac{10609}{200}$$

$\Rightarrow P_0, P_1, P_2, \dots, P_n$

$P_0 = 50$

$$r = \frac{P_1}{P_0} = \frac{103/2}{50}$$

$r = 1.03$

By geometric sequence,
 $a_n = a_0 r^n$

$\Rightarrow \boxed{P_n = P_0 r^n}$ Ans

$\Rightarrow P_{25} = 50 \times r^{25}$

$P_{25} = 104.68$ million Ans people after
25 years.

(e)

$$P_k = \left(1 + \frac{i}{m}\right) P_{k-1}$$

$$P_1 = \left(1 + \frac{i}{m}\right) P_0$$

$$P_2 = \left(1 + \frac{i}{m}\right) P_1 = \left(1 + \frac{i}{m}\right) \left(1 + \frac{i}{m}\right) P_0 = \left(1 + \frac{i}{m}\right)^2 P_0$$

$$P_3 = \left(1 + \frac{i}{m}\right) P_2 = \left(1 + \frac{i}{m}\right)^3 P_0$$

$$\Rightarrow \boxed{P_n = \left(1 + \frac{i}{m}\right)^n P_0} \rightarrow \text{Explicit formula}$$

We can see the common ratio is $\left(1 + \frac{i}{m}\right)$ &
this is a geometric sequence

Q 5:

(a) Bitstrings of length 10:

(i)

Exactly 4 1's = ?

4 1's are chosen from 10

$$= {}^{10}C_4 = \frac{10!}{4!(10-4)!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot \cancel{6!}}{1 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot \cancel{6!}} = 210 \quad \underline{\text{Ans.}}$$

(ii)

Atmost 4 1's = ?

= (B-str with 0 1's) + (B-str with 1 1's) + ... (B-str with 4 1's)

$$= {}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4$$

$$= 386 \quad \underline{\text{Ans.}}$$

(iii)

Atleast 4 1's = ?

= (Total bitstrings) - (B-str with atmost 3 1's)

$$= 2^{10} - ({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3)$$

$$= 848 \quad \underline{\text{Ans.}}$$

(b)

Total Q's = 40

True = 17

False = 40 - 17 = 23

No. of ans keys = ?

$$= {}^{40}C_{17} = {}^{40}C_{23}$$

$$= 8.87 \times 10^{10} \quad \underline{\text{Ans.}}$$

Either choose ~~ans~~ answer keys of true answers of false.

(c) Licence plates = ?

(i)

No letter repeat:

Letters Digits

$$\begin{aligned}\text{Letters} &= {}^{26}C_1 \cdot {}^{25}C_1 \cdot {}^{24}C_1 \\ &= 15600\end{aligned}$$

(one letter reduced so it stays unique)

$$\begin{aligned}\text{Digits} &= {}^{10}C_1 \cdot {}^9C_1 \cdot {}^8C_1 \\ &= 720\end{aligned}$$

$$\begin{aligned}\text{Total licence plates} &= \text{Letters} \times \text{Digits} \\ &= 15600 \times 720 \\ &= 11.232 \times 10^6, \text{ Ans.}\end{aligned}$$

(ii)

Letter can repeat:

Letters Digits

$$\begin{aligned}\text{Letters} &= {}^{26}C_1 \cdot {}^{26}C_1 \cdot {}^{26}C_1 \\ &= 17576\end{aligned}$$

$$\text{Digits} = {}^{10}C_1 \cdot {}^{10}C_1 \cdot {}^{10}C_1 = 1000$$

$$\begin{aligned}\text{Total license plates} &= \text{Letters} \times \text{Digits} \\ &= 17.576 \times 10^6, \text{ Ans.}\end{aligned}$$

Q 3 :

1. Trace table: $A=14$, $B=19$

$y \neq 0?$	x	y	$r (r = y \bmod 2)$	$r = 0?$	$r = 1?$	Product	Step
	14	19				0	Initial
T	14	18	1	F	T	14	1
T	28	9	0	T	F	14	2
T	28	8	1	F	T	42	3
T	56	4	0	T	F	42	4
T	112	2	0	T	F	42	5
T	224	1	0	T	F	42	6
T	224	0	1	F	T	266	7
F	-	-	-	-	-	-	8

$\Rightarrow$ Product = 266

2. Proof:

• Pre-conditions:

A & B are positive integers, $x = A$, $y = B$,
product = 0

• Basis step:

$I(0): AB + 0 = AB$

$AB = AB \quad \checkmark$

• Inductive Step: $I(n) \rightarrow I(n+1)$

Inductive hypothesis:

$I(n): \underline{xy + \text{product} = AB}$ ^① is true for n^{th} iterations

To show: $I(n+1)$ is also true for n^{th} iterations

$I(n+1):$

If $r = 0$,

$$\textcircled{1} \Rightarrow (2x) \left(\frac{y}{2} \right) + \text{product} = AB$$

$\Rightarrow \underline{xy + \text{product} = AB}$, which is true by inductive hypothesis

If $r=1$,

$$\textcircled{1} \Rightarrow x(y-1) + (\text{product} + x) = AB$$

$$\Rightarrow xy - \cancel{x} + \text{product} + \cancel{x} = AB$$

$$\Rightarrow \underline{xy + \text{product} = AB}$$
, which is true

So, $I(n+1)$ is also true for n^{th} iterations.

• Guard Condition:

$$\Rightarrow \underline{y \neq 0}$$

From the trace table and observation of program, we can see that y is either divided by itself or decremented by 1. So, eventually when it will divide by itself we will get 1 & the decrementing in next step will give 0, so guard condition will be met.

• Post-Conditions:

After the guard condition is met, $\underline{y=0}$,

$$\text{so, } \textcircled{1} \Rightarrow x(0) + \text{product} = AB$$

$$\boxed{\text{product} = AB} \text{ which is the post-condition}$$

So, the algorithm is correct. ✓